

How structured is a graphon?

Motivation

Let $f : [0, 1]^2 \rightarrow [0, 1]$ be a graphon, ie a symmetric function. Traditionally, we generate random graphs using f using the following algorithm. Let

$$X_1, \dots, X_n \sim \text{Uniform}([0, 1]),$$

G_n be a random graph with n vertices where

$$\{(v_i, v_j) \in E(G_n)\} \sim \text{Bernoulli}(f(X_i, X_j)).$$

From this perspective, locally it seems that $f(x, y) = p$ or Erdős Renyi Graphs have the highest entropy. But if you look at the graph from a global perspective, you see 1 community as $n \rightarrow \infty$ and thus Erdős Renyi Graphs end up having the most structure. Something similar happens with block stochastic graphons which correspond to multipartite-esque graphs (n “communities” with maybe a few connections between communities). So it makes sense to study graphons and the sequences that they produce from a more global perspective.

The impetus for the problem comes from the following idea, what if f , a graphon doesn't describe probability of an edge but rather edge weight of a complete graph. Let $G_n = (K_n, \Phi)$ where $\Phi : E(K_n) \rightarrow [0, 1]$ where

$$\varphi(v_i, v_j) = f(X_i, X_j).$$

Let $(G_n)_{n \in \mathbb{N}}$ be the sequence of random weighted complete graphs generated in this way. Note two graphs are equal if they are results of a simple permutation of vertices. Let H_f be the set of all possible complete weighted graphs that can be produced by this algorithm. You will notice a few things:

1. If $f = p$, then $|H_f| = 1$.
2. If f is a block graphon with n blocks, then $|H_f| = n$
3. The less structured the graphon is the more sequences are possible. Note you start to care about asymptotic growth.
4. At the extreme, a graphon $f(x, y) \sim \text{Uniform}([0, 1])$ should have the most sequences

One can logically ask the question how many possible sequences of complete weighted graphs can you create from such a process for a given graphon f . This should be equivalent to asking the question how complicated is the graphon f .

Problem

Can you quantify how random a graphon is or how much entropy does a graphon have?

Possible Implications

- Sampling Bounds: If you can quantify how complex a graphon is you may be able get learning-theoretic results. For a expected model with lower complexity, you may need fewer samples.
- Understand Larger Network Structure: Real world networks typically exhibit a spectrum of structure:
 - Social networks: community structure (low-medium complexity)
 - Biological networks: some motif structure but noisy (medium complexity)
 - Random graphs for null models: high complexity
- Compression and Information Theory: Descriptive complexity = compressibility

- Algorithm Design and Computational Complexity: The more global structure can aid algorithmic tasks. For example community detection is fast on block models, hard on random graphs. Graph matching is easier with structure

You can get a spectrum of complexity.

Prediction 1: Community detection should work better on graphs sampled from low-complexity graphons

Test: Sample from varying complexity graphons, measure detection accuracy

Prediction 2: Real networks with “good” community structure should have low estimated graphon complexity

Test: Estimate graphon from real networks, measure complexity Hypothesis: Social networks (known communities) $\rightarrow$ low complexity Hypothesis: Random ER graphs $\rightarrow$ high complexity

Prediction 3: Complexity should predict limits of community detection

Information-theoretic threshold: can you detect communities? Computational threshold: can you efficiently detect them? Both should relate to graphon complexity

Maybe you can give confidence bounds given a sequence of graphs on community detection.

Possible related problems

What if you sample f from a probability distribution?

Ideas

- Szemerédi’s Lemma
- Sobolev Norm
- Kolmogorov Entropy